

Lab 1 – Code Description

User.java

The `User` class represents a single user with a username and password. When a user is created, both fields are validated immediately — the username must be a properly formatted email address no longer than 50 characters, and the password must be between 8 and 12 characters and contain at least one letter, one digit, and one special character from an allowed set. If any rule is broken, an exception is thrown and the object is not created. The class also supports sorting by implementing `Comparable`, ordering users by their username's ASCII value.

UsersApp.java

`UsersApp` is the main entry point of the program. It reads a file whose name is passed as a command-line argument, processes each line as a username–password pair, and tries to create a `User` object for each one. Lines with extra spaces are handled correctly, and any validation error is printed to the console without stopping the rest of the file from being processed. Once all lines are read, the valid users are sorted by ASCII value and printed to the console.